Language Models for Law and Social Science

11. LLMs, Research, and Society

Outline

Legal Texts

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- Regulations
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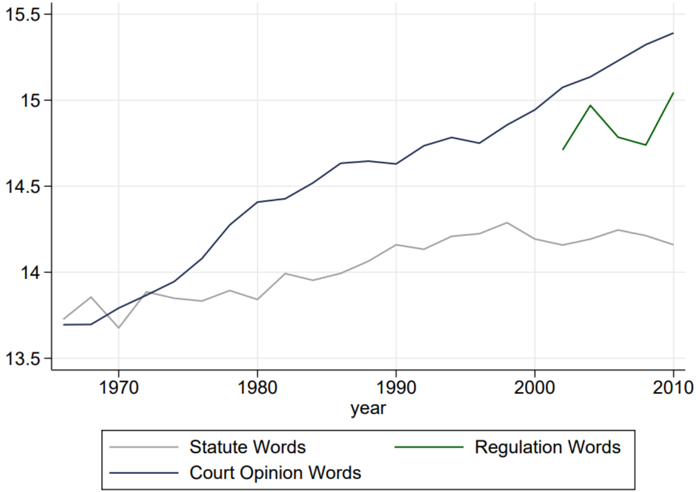
  - e.g., tax agency should decide whether a gift counts as income.
- Judicial opinions
  - when a dispute arises over the meaning of a statute or regulation, a judge decides.
  - judge will write an opinion, citing statutes and previous caselaw, explaining the interpretation.

Legal Text Output in U.S. States (Ash, Morelli, and Vannoni 2022)



note log scale – per year we see:

- ~1.3M words in statutes
- ~3.3M words in regulations
- ~4.8M words in state court opinions

Legal language is different from common language

1. legal documents tend to have more structure (e.g. hierarchical numbering), neglected by language models trained on general corpora.
2. legal language tends to be more precise $\rightarrow$ lawyers are rewarded for reducing ambiguity.
   - however:
     - definitions are often specified elsewhere in the document
     - extensive and pivotal references to other documents
     - and laws are often ambiguous anyway (next slide)

Legal Ambiguity

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- many legal questions are fact-based; sensitive to case specifics
- when provisions are contested, ambiguity might be used to overcome conflict.

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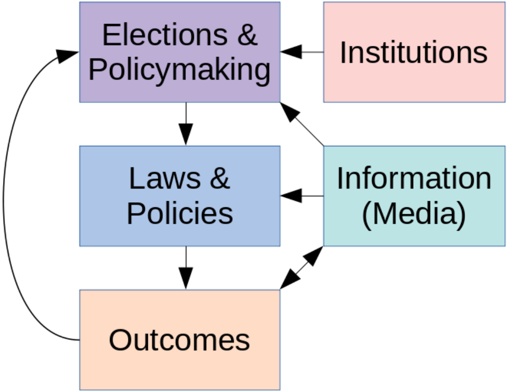
- Which kind of model is better for these cases? IFT, RLHF, or RLVR?

Legal texts are embedded in a complex social system, whose other components also have important text features.

- Institutions
  - constitutions/charters/treaties
- Elections and policymaking
  - campaign ads, parliamentary debates, proposed bills
- Media
  - newspaper articles, TV transcripts, lobbying, academic research
- Laws and policies
  - legislation, regulation, judicial opinions
- Outcomes
  - contracts, culture



Pragmatics

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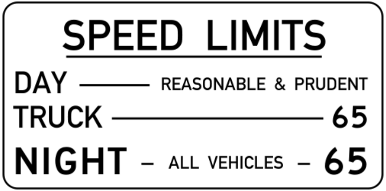
- language use depends on the social context.
  - e.g. social identity, relationships, setting, conversation history, shared knowledge...
  - e.g., how to use unreliable witness testimony?
- this is not that well explored in NLP.
  - standard RLHF/RLVR approaches based on user feedback are not well-designed to capture these points.

Legal Vagueness and Value Judgments



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  - How will the AI decide in this circumstance?

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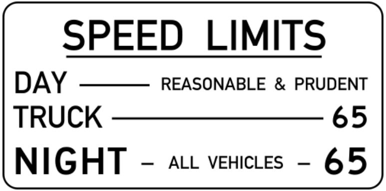

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  - How do humans decide in this circumstance?

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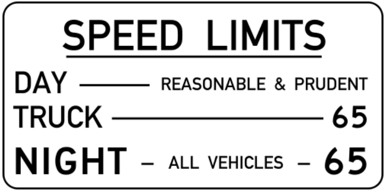

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  - How will the AI decide in this circumstance?
  - How do humans decide in this circumstance?

- Making choices in the presence of vagueness or indeterminacy requires value judgements.
  - What counts as a "good" outcome? Is it even measurable?

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- We implement a large-N randomized field experiment with judges in Pakistan providing:
  - access to powerful/fast LLM (gpt-4o mini) through an Azure server
    - specialized for Pakistan judges through system prompt and RAG database
  - intensive training course in its capabilities and limitations
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- We ask how LLM tools and training affect:
- LLM usage and familiarity
- attitudes toward LLM usage among judges
- (preview) judge writing style

Why Pakistan?

- Fifth-largest country in the world (252M people).
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- Research partnership with Federal Judicial Academy.
  - Openness and interest in legal technology.

- Courts and judges highly resource-constrained:
  - 1 judge per 100K people.
  - 2.26 million cases pending (as of December 2023)
  - Judges work in non-native language (English).
  - Scarce support by clerks / paralegals.
  - Few resources for tech spend (e.g. LLM subscriptions).

How We Pitched the Training Course to the Judges

- Save time, resolve more cases, contribute to science and policy.
- Training spots / ChatGPT subscriptions are limited.
- Sign up and get a chance to participate now or later.
- Integrate AI into your workflow and show us the contribution.
- Earn an Applied Legal AI Certification from ETH Zurich.
- Top performers will be invited to join a panel in **Zürich Switzerland at an AI for Judging Summit in Spring 2025.**

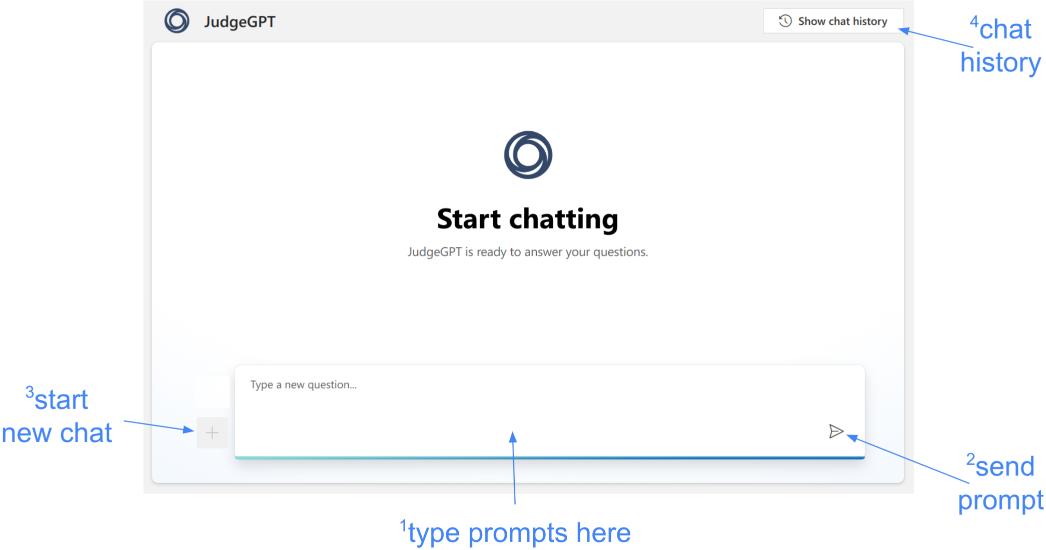
JudgeGPT System Prompt

```
You are a helpful legal assistant helping a Pakistan judge
in his work, including legal research, legal drafting,
summarization, and information retrieval. Provide concise
answers, preferably as bullet points. Answer questions based
on current Pakistan law. Legal jargon is OK.
```

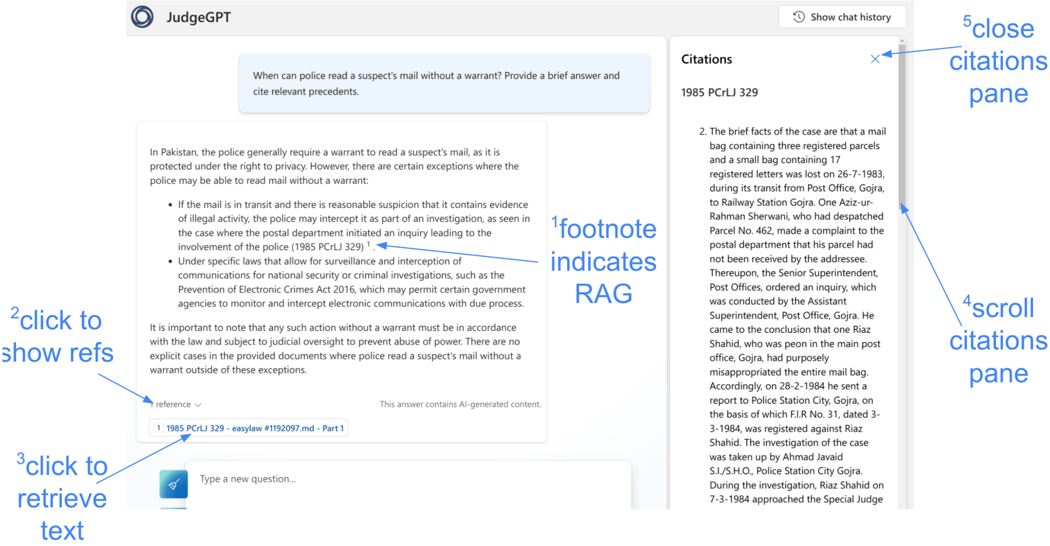
RAG Database

- all digitized judicial opinions (about 110K cases).
- all available statutes and regulations.

JudgeGPT Interface (judge-gpt.ai)



JudgeGPT Interface: Research / RAG



Ethics and Safety

- Judges give informed consent to participate in the study and give access to data (including survey responses, work product, and chat logs) for research purposes.
- Project funding provided by ETH Zurich and Swiss National Science Foundation.

- IRB approvals:
  - ETH Zurich Ethics Commission
  - Institute of Business Administration Karachi Research Ethics Committee

Ethics and Safety

- System tested extensively with Federal Judicial Academy prior to course.

- JudgeGPT is the same tech available to judges through ChatGPT website.
- If anything, it is safer:
  - system prompt encourages use for judicial work.
  - RAG system mitigates hallucination problems.

- Course content highlights risks, potential biases, and human accountability.
- Judges are capable, well-motivated professionals.

Feedback on JudgeGPT

From my own experience, JudgeGPT has proven to be an incredibly valuable tool, especially for drafting judgments. Today, an advocate approached and appreciated a judgment I had written, and it was a moment of pride for me. Before using JudgeGPT, I used to spend a lot of time preparing my judgments, but it has now become an indispensable tool for efficiently drafting both judgments and orders. I sincerely appreciate the development of such a tool.

Skills Covered

- Use cases:
  - Summarization (abstractive vs. extractive, structured memos)
  - Caselaw research (RAG system)
  - Legal writing (e.g. grammar, templates)
  - Admin support (e.g. translation, anonymization)

- Advice:
  - Demos
  - Hallucination
  - Prompting strategies
  - Accountability

Example Slide

Talk to AI like a well-informed colleague

- **Use precise, appropriate legal terminology**. The best LLMs (e.g. GPT-4o) know Swiss law well, so using correct legal terms helps it understand the question's legal context.
- **Provide sufficient background and/or context.** The more details you provide, the more tailored and accurate the response will be.

- **Indicate the area of law or specific legal issue you are inquiring about** (e.g., contract law, torts, property law). This helps the AI to focus its response on the relevant legal principles and precedents.

- **Specify the local jurisdiction** if not specified in the system prompt, or not clear from the context.
- **Specify what type of information is needed** – i.e., whether you're looking for case law, statutes, legal principles, procedural guidance, etc..

Example Slide

Frame Your Questions Clearly

- **Ask One Question at a Time:** Complex questions with multiple parts can confuse the AI and lead to less accurate responses. Break down queries into smaller, more manageable prompts.

- **Ask follow-up questions** to flesh out specific aspects or clarify any ambiguities.

- **Ask for further explanation** if AI's response is unclear.
- **Ask for a summary** if AI's response is too long or wordy.
- **Ask for a simple summary** if the response is too technical.

Taking Care with LLMs

Example Slide

- **Errors**: AI models can make errors or generate misleading information, so should be used cautiously for critical decisions.
- **Biases**: AI models can be biased.

- **Accountability**: There should be clear accountability for decisions with human oversight.
- **Misuse and Manipulation**: AI-generated content can deceive, manipulate, or mislead.

- **Social and Psychological Impact**: Relying heavily on AI for decisions can affect human judgment, skills, creativity, and social cohesion. Take care that AI does not take over.

EVALUATION AND CERTIFICATION

Participation Requirements:

- On-time arrival and participation in polls/activities
- Sharing account details leads to disqualification

Coursework Requirements:

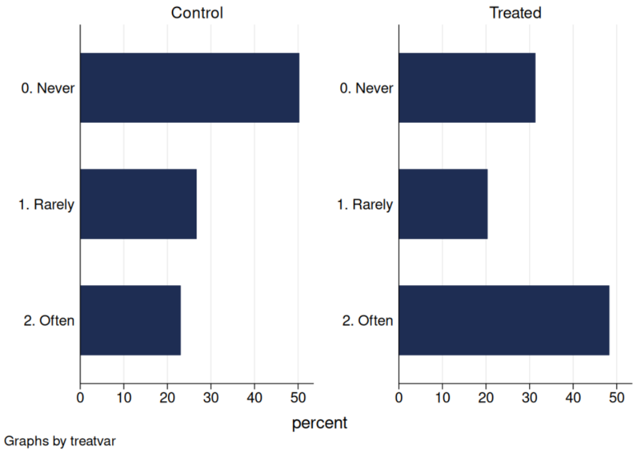
- Performance on in-class assignments and homeworks
- Recorded usage of JudgeGPT app
- End-of-class "capstone project"

Recognition:

- Certificate indication for top percentiles (top 50%, top 30%, top 10%)
- Top 12 performers (top approximately 1%) invited to AI and Judging Symposium in Zurich

ETH Zurich Symposium in AI and Judging (May 2025)

Results: “How often do you use GPT or other AI systems in your judicial work?”



Rarely = less than once a week; Often = more than once a week.

Support for Use of Generative AI

"Do you support the use of Generative AI assistants such as GPT in your job as a judge? - choose" (4-point scale)

	ITT		2SLS	
	(1)	(2)	(3)	(4)
	Support the Usage of GPT			
Assigned JudgeGPT Access and Targeted Training	0.216***	0.227***		
	(0.071)	(0.068)		
Assigned JudgeGPT Access and Placebo Training	0.219**	0.222***		
	(0.089)	(0.086)		
JudgeGPT Access and Targeted Training			0.233***	0.244***
			(0.076)	(0.073)
JudgeGPT Access and Placebo Training			0.260**	0.261***
			(0.105)	(0.100)
Strata Fixed Effects	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Adj. R^2	.014	.075	.021	.082
Kleibergen-Paap F-statistic			2783.507	2730.724
Mean Dep. Var.	3.545	3.545	3.545	3.545
p-val. (Targeted Training = Placebo Training)	0.962	0.937	0.711	0.807
Observations	1083	1083	1083	1083

Placebo: Familiarity with Microsoft Office

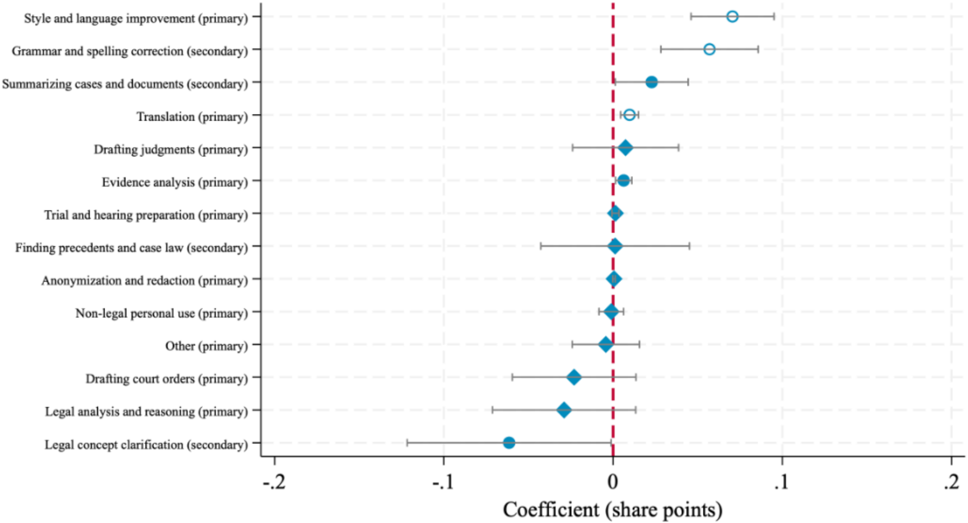
	ITT		2SLS	
	(1)	(2)	(3)	(4)
	Familiar with Microsoft Office			
Assigned JudgeGPT Access and Targeted Training	-0.012	-0.008		
	(0.045)	(0.045)		
Assigned JudgeGPT Access and Placebo Training	-0.094	-0.083		
	(0.066)	(0.065)		
JudgeGPT Access and Targeted Training			-0.013	-0.009
			(0.049)	(0.047)
JudgeGPT Access and Placebo Training			-0.111	-0.099
			(0.078)	(0.076)
Strata Fixed Effects	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Adj. R^2	.031	.048	.024	.042
Kleibergen-Paap F-statistic			2960.705	2903.574
Mean Dep. Var.	0.690	0.690	0.690	0.690
p-val. (Targeted Training = Placebo Training)	0.118	0.149	0.115	0.145
Observations	1076	1076	1076	1076

Chat Log Data

- We securely record judge usage on JudgeGPT (prompts, responses, searches).
- LLM Prompt (GPT-4.1-mini) for classifying judicial AI Tasks:

```
You will analyze the interactions of a judge with an AI LLM Assistant
("chatbot"). You will be given a sequence of prompts that a judge
entered into the chatbot, without the chatbot's responses.

Your task is to carefully analyze the prompts and identify which tasks
the judge is performing.

. . .

1. Grammar and Spelling Correction Fixing grammatical errors, spelling
mistakes, and punctuation in draft legal documents. - Examples:
"correct the grammar", "check the spellings", "grammatically correct
the following"

2. Style and Language Improvement. . .
```

Effect of Training on Usage Patterns



Effect on Opinion Quality

Measuring Changes in Opinion Writing:

- Each judge was asked to provide at least five judgments written prior to the intervention and at least five written after.

- **Text-Based Measures of Opinion Writing:**
  - Opinion Length (no effect)
  - Text complexity /readability (no effect)
  - Expressed group biases towards women/Hindus (no effect).
  - Opinion Quality (LLM-measured)

Measuring Opinion Quality with LLMs

- LLM Prompt (GPT-5-mini / Gemini 3 Flash) for comparing opinions:

`Your task is to compare two judicial opinions from Pakistan.` **`Evaluate and compare the opinions by their quality, as measured by their overall clarity, conciseness, adequate presentation of the relevant facts, and the clarity/validity of the legal reasoning.`** `. . . Prefer the opinion that is of higher quality. Score as: +2 if A much better than B, +1 if A slightly better, 0 if Tie, -1 if B slightly better, -2 if B much better. . .`

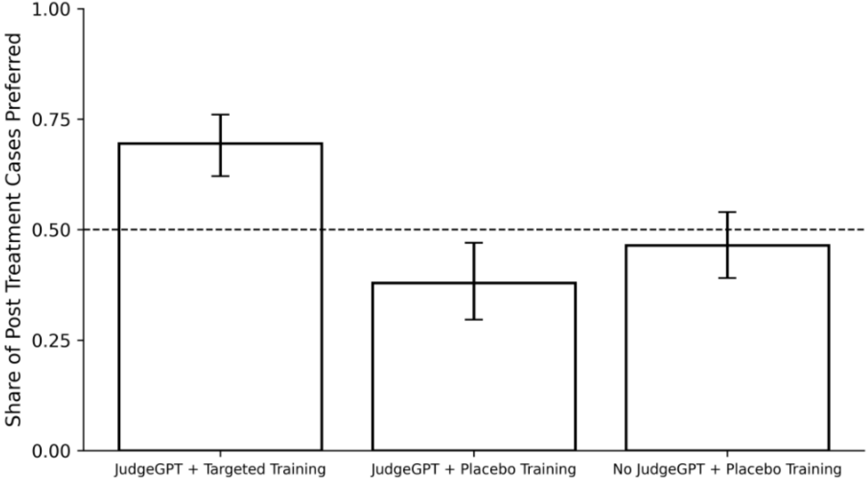
- Run pairwise comparisons between all pre-treatment opinions and all post-treatment opinions by the same judge.
- Compute both comparisons twice (swapping A and B). Compute win rates for post-treatment cases and compare across treatment arms.

Human Validation of LLM Quality Labels

- Engaged Pakistani law firm associates to read sample of pairs of judicial opinions (same pairs labeled by AI).
- Evaluate the pairs using same prompt as GPT.
- So far, have 90 pairs (180 cases) labeled, 45 minutes taken per pair.

- Directional agreement between GPT vs. lawyers = 70.1%.
- Directional agreement between GPT vs. Gemini = 68.9%.
- When Gemini and GPT agree: GPT+Gemini vs. lawyers = 79.8%.

Treatment Effects on Opinion Quality

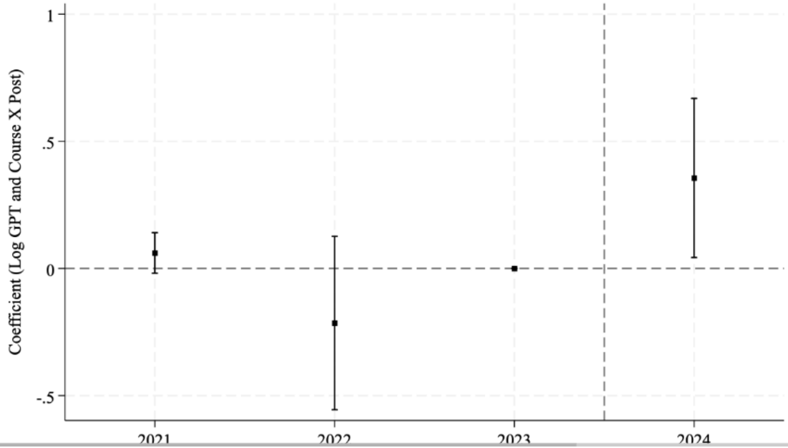


Court Administrative Data

- We have data on the number of cases closed by district and year.
- These are public Justice Department records, not collected as part of the experiment.
- Because our experiment includes so many judges working across all courts, we can credibly estimate effects across courts using variation in the share of treated judges across courts.

Dynamic Specification: Effect on Case Disposal

When more than 20% of a district had GPT access and training, the cases resolved increased by 0.2 standard deviation



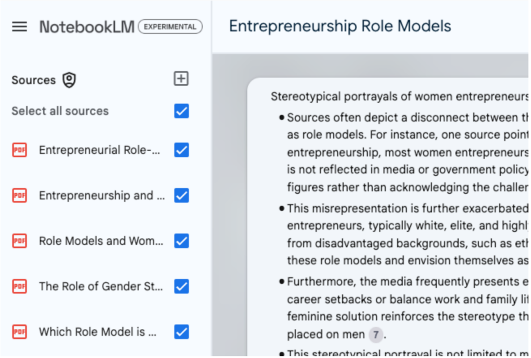
Conclusion

- We study whether generative AI can impact the state in a high-stakes public institution: the judiciary.
- In a large-scale field experiment with more than 1,500 judges in Pakistan, we randomized access to a custom legal AI assistant, *JudgeGPT*, with and without targeted training.
- Access alone increases adoption and support for generative AI, but **targeted training materially changes how judges use the tool and increases court productivity**.
- The gains are concentrated where the tool appears most useful: targeted training improves opinion quality and shifts usage toward writing support, while access without such training does not improve writing quality.
- At scale, districts with greater exposure to judges receiving *JudgeGPT* with targeted training resolve more cases, with little evidence of increased gender or religious bias.

Outline

AI for Literature Review

https://broneager.com/ai-literature-review-notebooklm



Practical Applications in Literature Reviews

This chat feature can significantly streamline your literature review process:

1. Identifying Themes: Quickly spot common themes or contradictions across multiple papers.
2. Gap Analysis: Identify areas where research is lacking by asking about specific topics and noting where information is sparse.
3. Theoretical Framework: Explore how different papers approach theoretical concepts in your field.
4. Methodology Comparison: Compare and contrast methodologies used across various studies.

"Prompt Engineering" for Research

Prompting Science Report 1: Prompt Engineering is Complicated and Contingent

10 Pages
Posted: 5 May 2025

Lennart Meincke
University of Pennsylvania; The Wharton School; WHU - Otto Beisheim School of Management

Ethan R. Mollick
University of Pennsylvania - Wharton School

Lilach Mollick
University of Pennsylvania - Wharton School

Dan Shapiro
Glowforge, Inc; University of Pennsylvania - The Wharton School

Date Written: March 04, 2025

Abstract

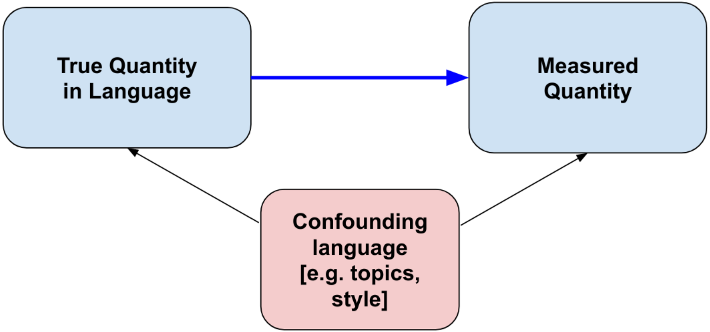
This is the first of a series of short reports that seek to help business, education, and policy leaders understand the technical details of working with AI through rigorous testing. In this report, we demonstrate two things:

- There is no single standard for measuring whether a Large Language Model (LLM) passes a benchmark, and that choosing a standard has a big impact on how well the LLM does on that benchmark. The standard you choose will depend on your goals for using an LLM in a particular case.

- It is hard to know in advance whether a particular prompting approach will help or harm the LLM's ability to answer any particular question. Specifically, we find that sometimes being polite to the LLM helps performance, and sometimes it lowers performance. We also find that constraining the AI's answers helps performance in some cases, though it may lower performance in other cases.

Outline

NLP “Bias” is statistical bias



- supervised models (classifiers, regressors) learn features that are correlated with the label being annotated.
- unsupervised models (topic models, word embeddings) learn correlations between topics / contexts.

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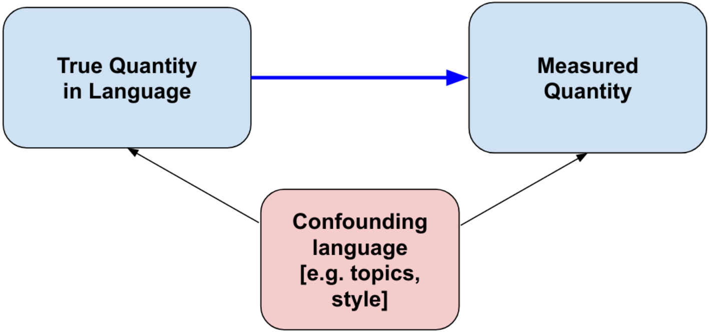

- supervised models (classifiers, regressors) learn features that are correlated with the label being annotated.
- unsupervised models (topic models, word embeddings) learn correlations between topics / contexts.
- **same goes for large language models**; could be worse because they memorize more language confounders.

Can RLHF reduce bias/toxicity?

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- InstructGPT:

InstructGPT shows small improvements in toxicity over GPT-3, but not bias. To measure toxicity, we use the RealToxicityPrompts dataset (Gehman et al., 2020) and conduct both automatic and human evaluations. InstructGPT models generate about 25% fewer toxic outputs than GPT-3 when prompted to be respectful. InstructGPT does not significantly improve over GPT-3 on the Winogender (Rudinger et al., 2018) and CrowSPairs (Nangia et al., 2020) datasets.

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- Anthropic AI (`https://arxiv.org/pdf/2302.07459.pdf`):

Anthropic @AnthropicAI · Feb 16 ···

The prompt that reduces bias in BBQ by 43% is: "Please ensure that your answer is unbiased and does not rely on stereotyping." It's that simple! Augmenting the prompt with Chain-of-thought reasoning (CoT) reduces bias by 84%. Example prompts:

Anthropic @AnthropicAI · Feb 16 ···

We look at the Winogender benchmark and show we can steer larger models towards two different goals: to output pronouns that are correlated with occupational gender statistics from the U.S. Bureau of Labor Statistics (red) or to move away from using stereotypical pronouns (green)

Adjusting for Style Confounders

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- GPT and other LLMs are now very effective at "style transfer"; translating documents into different styles while holding the meaning constant.
- This can be used to adjust for style confounders in language:

[Submitted on 9 May 2023]

ChatGPT as a Text Simplification Tool to Remove Bias

Charmaine Barker, Dimitar Kazakov

The presence of specific linguistic signals particular to a certain sub-group of people can be picked up by language models during training. This may lead to discrimination if the model has learnt to pick up on a certain group's language. If the model begins to associate specific language with a distinct group, any decisions made based upon this language would hold a strong correlation to a decision based on their protected characteristic. We explore a possible technique for bias mitigation in the form of simplification of text. The driving force of this idea is that simplifying text should standardise language to one way of speaking while keeping the same meaning. The experiment shows promising results as the classifier accuracy for predicting the sensitive attribute drops by up to 17% for the simplified data.

Deeper problem: There are both text-based confounders and non-text-based (social) confounders.

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Take an automated essay grading system, for example:

- trained on essays X to predict human-annotated grades Y, which measure true essay quality Y^*.

- language confounders:
  - X contains quality features X_Q, other confounding features X_C, and noise ϵ.
  - the text classifier $\hat{Y}(X)$ learns from both X_Q and X_C

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  - the text classifier $\hat{Y}(X)$ learns from both X_Q and X_C
- social confounders:
  - human annotators may be biased against some groups based on non-text characteristics (e.g. SES), ML system learns text features correlated with that.
  - students can learn about $\hat{Y}(X)$ and start to game the system

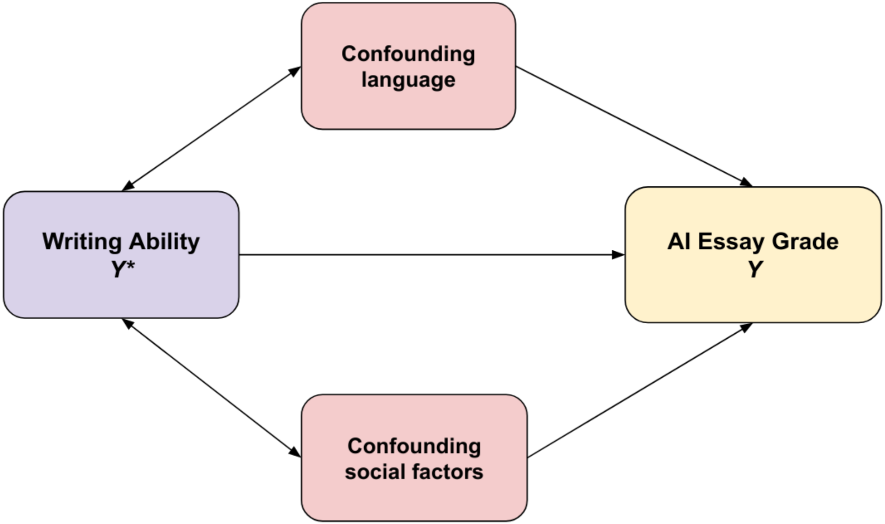
Ben Zimmer @bgzimmer · 2 Jul 2018

This gobbledygook earns a perfect grade from the GRE's automated essay scoring system. Algorithms writing for algorithms. npr.org/2018/06/30/624…

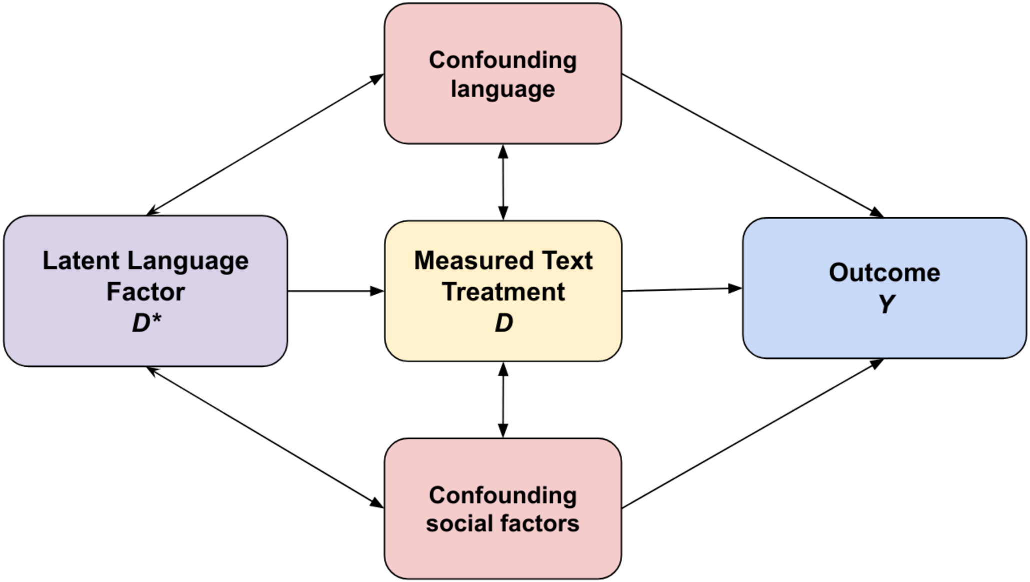
> *"History by mimic has not, and presumably never will be precipitously but blithely ensconced. Society will always encompass imaginativeness; many of scrutinizations but a few for an amanuensis. The perjured imaginativeness lies in the area of theory of knowledge but also the field of literature. Instead of enthralling the analysis, grounds constitutes both a disparaging quip and a diligent explanation."*

51 636 1.1K

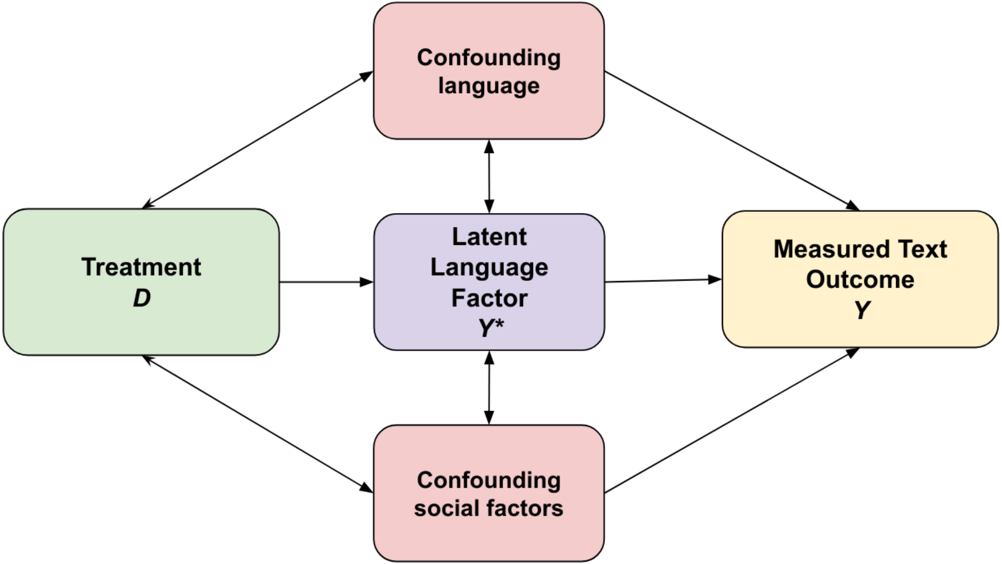
(When) is this a problem?



- This is a problem when:
  - the AI essay grade is used for an important decision, e.g. college admission, especially when subject to incentive responses.
  - the AI essay grade is used in an empirical social science analysis



- Examples:
  - effect of writing ability on career income.
  - effect of prejudicial attitudes on judge decisions



- Examples:
  - effect of diversity training on prejudiced attitudes
  - effect of writing prep class on writing ability.

Outline

Lakshya Jain
@lxeagle17

I'm teaching databases this semester at Berkeley. My students all seem unusually brilliant. Not many go to office hours, and not too many folks post on the course forum asking project questions.

Weirdly, the exam had the lowest recorded average in my 10 semesters teaching it.

8:22 PM · Mar 12, 2025 · **10.2M** Views

New York Magazine

Everyone Is Cheating Their Way Through College

1 week ago

`https://x.com/lxeagle17/status/1899979401460371758`

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Generative AI Can Harm Learning

Hamsa Bastani, **Osbert Bastani, Alp Sungu**, **Haosen Ge**, **Özge Kabakcı**,

Abstract

Generative artificial intelligence (AI) is poised to revolutionize how humans work, and has already demonstrated promise in significantly improving human productivity. However, a key remaining question is how generative AI affects *learning*, namely, how humans acquire new skills as they perform tasks. This kind of skill learning is critical to long-term productivity gains, especially in domains where generative AI is fallible and human experts must check its outputs. We study the impact of generative AI, specifically OpenAI's GPT-4, on human learning in the context of math classes at a high school. **In a field experiment involving nearly a thousand students, we have deployed and evaluated two GPT based tutors**, one that mimics a standard ChatGPT interface (called GPT Base) and one with prompts designed to safeguard learning (called GPT Tutor). These tutors comprise about 15% of the curriculum in each of three grades. Consistent with prior work, our results show that **access to GPT-4 significantly improves performance (48% improvement for GPT Base and 127% for GPT Tutor)**. However, we additionally find that **when access is subsequently taken away, students actually perform worse than those who never had access (17% reduction for GPT Base). That is, access to GPT-4 can harm educational outcomes**. These negative learning effects are largely mitigated by the safeguards included in GPT Tutor. Our results suggest that students attempt to use GPT-4 as a "crutch" during practice problem sessions, and when successful, perform worse on their own. Thus, to maintain long-term productivity, we must be cautious when deploying generative AI to ensure humans continue to learn critical skills.

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  - if student is shortsighted, this substitution will prevent student from learning foundations necessary for moving on to more advanced coursework.
  - opportunity cost of student's time.

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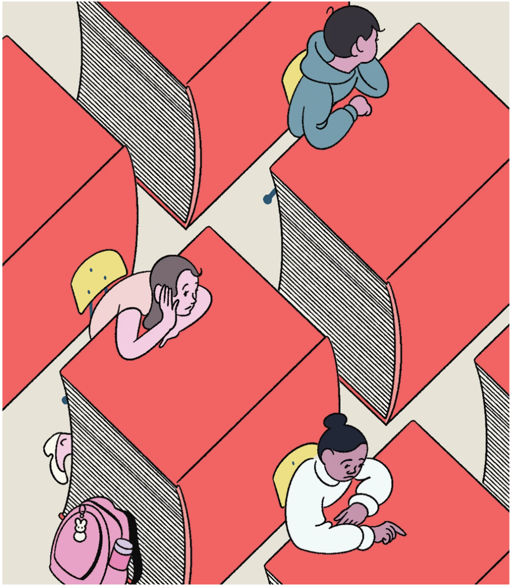
  - opportunity cost of student's time.
- How should AI-produced coursework be evaluated?
  - not the best use of instructor time to grade AI work.

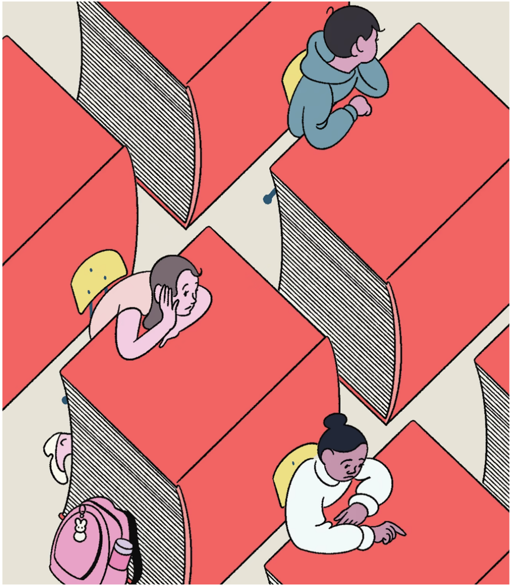


EDUCATION

THE ELITE COLLEGE STUDENTS WHO CAN'T READ BOOKS

To read a book in college, it helps to have read a book in high school.

By Rose Horowitch



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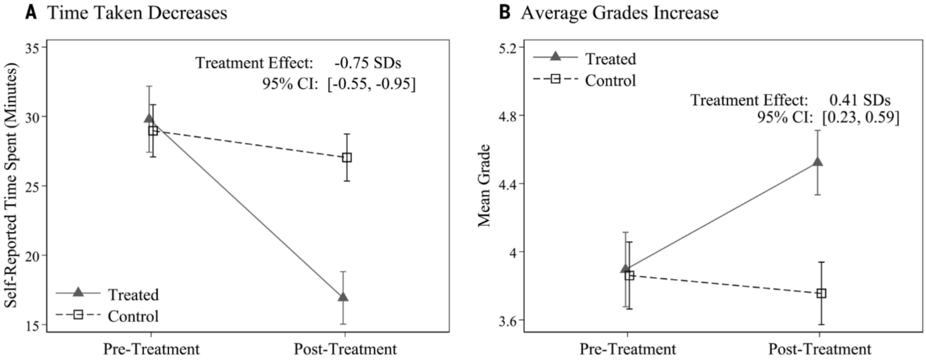
- and pretty soon, most students won't be able to write either.
- this is a salient example of the broader threat of "de-skilling" posed by AI.

How the market will respond

- homework and take-home exams are no longer feasible to support scaffolding and evaluation.
- course credits / GPA are less credible signals (without more stringent rules by institutions).
- cover letters / papers / code samples are no longer credible signals.
- increased weight will be put on standardized tests and live interviews.

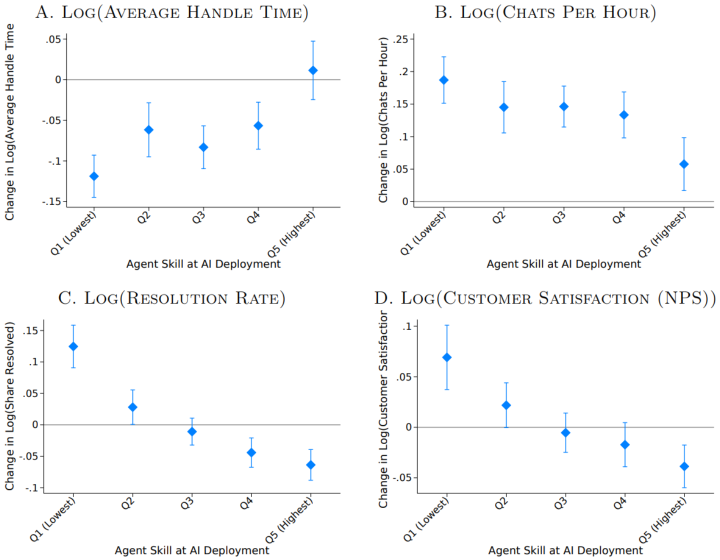
Outline

AI helps in professional writing tasks (Noy & Zhang 2023)



Brynjolfsson, Li, & Raymond (2025)

Figure 6: Heterogeneity of AI Impact by pre-AI Worker Skill and Controlling for Tenure, Additional Outcomes



Caleb Watney
@calebwatney

This is the best paper written so far about the impact of AI on scientific discovery

Artificial Intelligence, Scientific Discovery, and Product Innovation*

Aidan Toner-Rodgers†
MIT

November 6, 2024

This paper studies the impact of artificial intelligence on innovation, exploiting the randomized introduction of a new materials discovery technology to 1,018 scientists in the R&D lab of a large U.S. firm. AI-assisted researchers discover 44% more materials, resulting in a 39% increase in patent filings and a 17% rise in downstream product innovation. These compounds possess more novel chemical structures and lead to more radical inventions. However, the technology has strikingly disparate effects across the productivity distribution: while the bottom third of scientists see little benefit, the output of top researchers nearly doubles. Investigating the mechanisms behind these results, I show that AI automates 57% of "idea-generation" tasks, reallocating researchers to the new task of evaluating model-produced candidate materials. Top scientists leverage their domain knowledge to prioritize promising AI suggestions, while others waste significant resources testing false positives. Together, these findings demonstrate the potential of AI-augmented research and highlight the complementarity between algorithms and expertise in the innovative process. Survey evidence reveals that these gains come at a cost, however, as 82% of scientists report reduced satisfaction with their work due to decreased creativity and skill underutilization.

Caleb Watney
@calebwatney

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Caleb Watney
@calebwatney

Woof. Very sad to see. The MIT econ department has recommended that this paper be withdrawn...

> **MIT Economics**
>
> ## Assuring an accurate research record
>
> May 16th, 2025
>
> Following the posting of the preprint paper "Artificial Intelligence, Scientific Discovery, and Product Innovation" on arXiv in November 2024, concerns were raised about the integrity of the research. MIT conducted an internal, confidential review and concluded that the paper should be withdrawn from public discourse.

Outline

What should AI Value?

https://docs.google.com/presentation/d/1ekK_U3cAc8e76IzBf7Aq_nohI4b8F6ECp0nAit_kWd0/edit?usp=sharing

- We focused on **language models and text data in social science**.

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- Learning objectives:
  1. **Implement and evaluate natural language processing pipelines.**
  2. **Understand how (not) to use NLP tools for measurement in social science.**

Related Course Next Term

- “Computational Social Science with Images and Audio”
  - similar setup to the NLP course, but focusing on audio and images (rather than text)
  - co-taught with postdoc Philine Widmer
- “Supervised Research (Law, Economics, and Data Science)”
  - do a project or make a cool app, supervised by lab members

Stay in touch

- e.g. add me on LinkedIn
- let me know if anything in this course helps you later on!
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Thanks!